Randomized comparison of low-dose versus standard-dose praziquantel therapy in treatment of urinary tract morbidity due to Schistosoma haema tobium infection.

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At present, anthelmintic therapy with praziquantel at a dose of 40 mg/kg of body weight is the recommended treatment for control of urinary tract morbidity caused by Schistosoma haematobium. Although this standard regimen is effective, drug cost may represent a significant barrier to implementation of large-scale schistosomiasis control programs in developing areas. Previous comparison trials have established that low-dose (20-30 mg/kg) praziquantel regimens can effectively suppress the intensity of S. haematobium infection in endemic settings. However, the efficacy of these low-dose regimens in controlling infection-related morbidity has not been determined in a randomized field trial. The present random allocation study examined the relative efficacy of a 20 mg/kg dose versus a 40 mg/kg dose of praziquantel in control of hematuria and bladder and renal abnormalities associated with S. haematobium infection in an endemic area of Coast Province, Kenya. After a nine-month observation period, the results indicated an advantage to the standard 40 mg/kg praziquantel dose in terms of reduction of infection prevalence and hematuria after therapy ($P < 0.01$ and $P < 0.005$, respectively). However, the two treatment groups were equally effective in reducing structural urinary tract morbidity detected on ultrasound examination. We conclude that in certain settings, a 20 mg/kg dose of praziquantel may be sufficient in providing control of morbidity due to urinary schistosomiasis in population-based treatment programs.

Schistosomiasis drug therapy and treatment considerations.

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Schistosomiasis, a grave and debilitating disease of socioeconomic importance, is increasing in incidence despite concerted efforts to control and contain the disease in all the endemic areas. While a multipronged method of control using health education, sanitation and snail control has been used, chemotherapy and chemoprophylaxis play the most important and crucial role in containing/preventing the transmission of the disease. Schistosomicides such as antimonials were introduced, as early as the 1990s as the drugs of choice and continued to be used until the early 1960s. The antimonials were administered intravenously, and produced severe side effects; the various variables that determined their effects at the site of action made their application difficult and adversely affected their use in large scale chemotherapy. The antimonials were then replaced by hycanthone and lucanthone which were administered intramuscularly. These drugs produced immediate side effects such as hepatotoxicity and gastrointestinal disturbances, and were consequently withdrawn. It was then decided that the alternative was to produce synthetic drugs that could be administered orally. Niridazole, oxamniquine, and metrifonate were introduced as schistosomicidal agents, with drugs like oltipraz and amoscanate still at clinical trial phase. Therapeutic doses of drugs like hycanthone, niridazole and amoscanate have been found to cause many major side effects. A significant advance in the control of schistosomiasis chemotherapy is the introduction of a relatively safe, effective, broad spectrum oral helminthic agent, praziquantel. Studies have also shown that oxamniquine is as effective as praziquantel in eliminating intestinal S. mansoni infection, and metrifonate is as effective as praziquantel in eliminating urinary S. haematobium and S. mansoni infections. Praziquantel has been found to be effective in treating S. haematobium infections compared with metrifonate and more effective in treating S. mansoni infection when compared with oxamniquine. Because the drug is effective even when treating advanced hepatosplenic schistosomiasis, with few side effects, praziquantel is currently the drug of choice for the treatment of any kind of schistosomiasis. The only limitation is the cost which restricts its use in many developing countries. While effective, safe drugs for mass chemotherapy are being developed, the problem of therapeutic failure and drug resistance is being reported from certain developing countries. Under these circumstances, alternative drugs must be resorted to. Mass treatment, a crucial goal in the eventual control of schistosomiasis, awaits a well-tolerated and nontoxic drug that will ultimately prove to be effective where cure is definite. Until such a time, while eradication of the disease is a near impossibility, reducing the intensity of infection can ultimately reduce morbidity and even mortality.(ABSTRACT TRUNCATED AT 400 WORDS)

Therapeutic failure of praziquantel in the treatment of Schistosoma haematobium infection in Brazilians returning from Africa.

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Several cases of therapeutic failure of praziquantel used for the treatment of urinary schistosomiasis have been reported. Alternative drugs, like niridazol and metrifonate, have shown a lower therapeutic effect and more side effects than praziquantel. Twenty-six Brazilian military men (median age of 29 years) with a positive urine parasitological exam who were part of a United Nation peace mission in Mozambique in 1994 were treated with 40 mg/kg body weight praziquantel, single dose. They swam in Licungo river (Mocuba city, Mozambique) during the weekends. After this, they presented haematuria, dysuria, polakiuria, and lumbar pain. Control cystoscopy examinations carried out between 6 and 24 months after each treatment (including two additional treatments at a minimum interval of 6 months) revealed the presence of viable eggs. Granulomas in the vesical submucosa were observed in 46.2% (12/26) of the individuals. A vesical biopsy confirmed the presence of granulomas in all of these patients and the presence of viable eggs in 34.3% (9/26) of individuals who no longer excreted eggs in urine. The eggs filled with miracidia showed characteristics of viability. Histopathological examination using different strains demonstrated therapeutic failure and the need for repeated treatment. In this study, we demonstrated a low efficacy of praziquantel in the treatment of schistosomiasis haematobia, and the necessity of the urinary bladder biopsy as criterion of cure.

Schistosoma haematobium infections among schoolchildren in central Sudan one year after treatment with praziquantel.

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Chemotherapy with praziquantel (PZQ) is the mainstay of schistosomiasis control. However, there are recent concerns about tolerance or resistance to PZQ, so that monitoring its efficacy in different settings is required. A longitudinal study was conducted to evaluate the impact of PZQ for the treatment of Schistosoma haematobium infection among schoolchildren at Al Salamania, Central Sudan. Parasitological examinations for S. haematobium were performed in a cohort of schoolchildren (6-15 years of age) before and 1 year after treatment with a single dose of PZQ 40 mg/kg. Out of 562 (309 boys and 253 girls) schoolchildren recruited from three elementary schools, 420 completed one longitudinal dataset that comprised of data from two time points; baseline, and follow-up 1 year after treatment with a single dose of PZQ 40 mg/kg for S. haematobium infection. A single dose of PZQ significantly reduced the prevalence of S. haematobium infection by 83.3% (from 51.4% to 8.6%) and the geometric mean intensity of infection of positive individuals by 17.0% (from 87.7 to 72.8 eggs/10 ml of urine) 1 year after treatment. While there was no significant difference in the reduction of the prevalence of S. haematobium infection between the gender or age groups, there was a significantly higher reduction of intensity of S. haematobium infection among girls in comparison with boys. There was a significant reduction of S. haematobium infection 1 year after PZQ treatment in this setting.

Efficacy of praziquantel in the treatment of Schistosoma haematobium infection among school-age children in rural communities of Abeokuta, Nigeria.

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Chemotherapy with praziquantel (PZQ) has been the cornerstone of schistosomiasis control over the last two decades. Being the only available drug for the treatment of over 200 million people worldwide, continuous monitoring of PZQ efficacy under the pressure of widespread use is therefore advocated. The efficacy of taking two doses of oral PZQ for the treatment of Schistosoma haematobium was examined among school children in Nigeria. Urine specimens were collected from 350 school children and examined using the filtration technique. Blood was collected for packed cell volume (PCV) estimation, and the weight and height of each child were estimated. S. haematobium egg positive pupils were treated with two oral doses of PZQ at 40 mg/kg with a four-week interval in between. Drug efficacy was determined based on the egg reduction rate (ERR). Among 350 school children, 245 (70.0%) - of which 132 were males and 113 were females, with an age range of 4 to 15 years - were diagnosed with S. haematobium. All the 245 infected children received a single oral dose of 40 mg/kg PZQ twice with a four-week interval in between and were followed up for 12 weeks. At four, eight and twelve weeks post treatment, the ERR was 57.1%, 77.6% and 100%, respectively. The ERR was significantly higher among the children with a light infection compared to those with a heavy infection. One hundred and twenty-one children were egg negative at four weeks post treatment, among which 1 (6.3) and 120 (52.4%) had heavy and light infections, respectively. Following the second round of treatment, the cure rate at eight weeks and twelve weeks was 85.3% and 100%, respectively. This study demonstrated the efficacy of taking two doses of oral PZQ for the treatment of urinary schistosomiasis among school children in Nigeria.

Therapeutic and operational profiles of metrifonate and praziquantel in Schistosoma haematobium infection.

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A systematic analysis of the existing literature has been undertaken to compare the therapeutic and operational profiles of metrifonate (CAS 52-68-6), and praziquantel (CAS 55-268-74-1), two anti-schistosomal compounds. The criteria evaluated were therapeutic efficacy against Schistosoma haematobium and other helminths, impact on pathology commonly associated with S. haematobium infection, frequency, type and duration of adverse reactions, health risk associated with inadvertent overdosage, applicability and practicality of treatment in various medical settings, tolerance and resistance, pharmacological properties, toxicity and economic aspects. It is concluded that both medical and operational criteria indicate that praziquantel is superior to metrifonate for the treatment of schistosomiasis caused by S. haematobium. Since, compared to praziquantel, metrifonate has a number

of disadvantages, future antischistosomal chemotherapy can do without this drug.

Chemotherapy-based control of schistosomiasis haematobia. IV. Impact of repeated annual chemotherapy on prevalence and intensity of Schistosoma haematobium infection in an endemic area of Kenya.

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To determine the effect of repeated, annual, age-targeted therapy on prevalence and intensity of Schistosoma haematobium infection in an endemic area, we treated all available, infected, school-age children (n = 2, 493) in the Msambweni area of Coast Province, Kenya with a randomized protocol of oral metrifonate (10 mg/kg for three doses each year) or praziquantel therapy (40 mg/kg as a single dose each year) for a period of one to three years. During 1984-1987, 1, 101 children completed three years of therapy, 550 received two years, and 842 received a single year. Annual followup revealed significant long-term suppression of S. haematobium infection in the targeted school-age population. Both cross-sectional analysis and study of individual outcomes suggested maximal suppression of infection after two years of therapy. Suppression lasted more than two years after cessation of treatment, and was associated with reduced community transmission (gauged by decreased prevalence among new study entrants and decreasing negative-to-positive conversion on annual parasitologic examinations). Comparison of metrifonate and praziquantel outcomes indicated greater suppression of infection and longer infection-free intervals for some subgroups given praziquantel. We conclude that annual population-based therapy targeted to schoolchildren has direct and indirect beneficial effects for endemic communities. In some specific situations, repeat therapy may not suppress transmission, and reduced drug efficacy may be observed after one to three years, suggesting the need for additional non-drug control measures in highly endemic villages.

The association of steroids and schistosomicides in the treatment of experimental schistosomiasis.

PMID: 2515622 | Indexed in PubMed

The best therapeutic approach to acute schistosomiasis (Katayama fever) is still unsettled. In this paper we report a synergistic effect between schistosomicides and steroids in the treatment of the early stages of Schistosoma mansoni infection in the mouse. CBA mice infected with 150 S. mansoni cercariae were treated with oxamniquine or praziquantel and dexamethasone or prednisolone. The rate of parasite egg excretion by treated mice and appropriate controls was monitored, and the mice were perfused 43 d after infection for estimation of worm burdens and tissue egg densities. Mice treated with schistosomicides alone or with schistosomicides plus steroids had worm burdens of similar size. Significant reductions in egg counts were, however, recorded in faeces, and in the intestines and livers (with consequent reduction in liver pathology), of mice treated with schistosomicide and steroid, when compared to mice treated with schistosomicide alone or steroid alone. The apparent inhibition of fecundity of S. mansoni by combining these drugs has clear implications for treatment of the Katayama syndrome.

Chemotherapeutic approaches to schistosomes: current knowledge and outlook.

PMID: 12049454 | Indexed in PubMed

The chemotherapy of schistosomiasis has improved greatly in the last few decades. There are three drugs available: metrifonate (active against Schistosoma haematobium), oxamniquine (active against Schistosoma mansoni) as narrow spectrum drugs and praziquantel, which is effective against all important adult schistosome species and their immature stages, as the drug of choice.

Efficacy and safety of praziquantel plus artemisinin-based combinations versus praziquantel in the treatment of Kenyan children with Schistosoma mansoni infection: open-label, randomized, head-to-head, non-inferiority trial.

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Praziquantel alone is insufficient for the control of schistosomiasis due to poor efficacy against juvenile worms and increasing concerns about the risk of drug resistance. We compared the efficacy and safety of praziquantel combined with four different artemisinin-based combinations to praziquantel alone in treating Schistosoma mansoni infection in Kenyan children. In this randomized, open-label, five-arm, head-to-head, non-inferiority trial, children (aged 9-15 years) with S. mansoni infection according to duplicate Kato Katz thick smears from a stool sample in the Mwea irrigation scheme of central Kenya, were enrolled. Participants were randomly assigned (1:1:1:1:1) via a computer-generated block randomization procedure to receive a single oral dose of praziquantel (PZQ) (40 mg/kg/day) alone or in combination with a 3-day course (4 mg/kg of artesunate) of artesunate plus sulfalene-pyrimethamine (As + SP), artesunate plus amodiaquine (As + AQ), artesunate plus mefloquine (As + MQ) or dihydroartemisinin-piperaquine (DHAP). Laboratory technicians were masked to treatment allocation, but participants, clinicians, and study nurses were not. The primary outcomes were the cure rate and frequency of adverse events, which were assessed 6 weeks after treatment in the available case population using a per-protocol analysis. The non-inferiority margin was set at -10% for the risk difference in cure rates between combination therapy and PZQ alone. Between 12 September 2018 and 11 January 2019, 540 participants were assigned to receive PZQ alone (n = 108), PZQ plus As + SP (n = 108), PZQ plus As + AQ (n = 108), PZQ plus As + MQ (n = 108), or PZQ plus DHAP (n = 108). Primary outcome data were available for 523 (96.9%) participants. The cure rate was 82.5% (85/103) in PZQ alone, 81.7% (85/104) in PZQ plus As + SP, 76.2% (80/105) in PZQ plus As + AQ, 88.7% (94/106) in PZQ plus As + MQ, and 85.7% (90/105) in PZQ plus DHAP arm. Non-inferiority was declared for PZQ plus As + MQ (difference 6.2 [95% confidence interval: -3.3 to 15.6]) and PZQ plus DHAP (3.2 [-6.7 to 13.1]) but not for PZQ plus As + SP (-0.8 [-11.2 to 9.6]) or PZQ plus As + AQ (-6.3 [-17.3 to 4.6]). Adverse events were reported by 26% (138/540) of participants, including abdominal pain, headache, and vomiting. There were no serious adverse events. Alternatives to praziquantel should include praziquantel plus artesunate-mefloquine or praziquantel plus dihydroartemisinin-piperaquine. However, further multicentre trials are needed in different epidemiological settings and population groups to confirm these findings.CLINICAL TRIALSThis study is registered with the Pan-African Clinical Trials Registry under PACTR202001919442161.
